

TABLE I
Device Setting

Display	Description	Function of switches
LCD backlight? :OFF	Set the LCD backlight on/off	• SET/EXIT—exit the setting menu • PLUS—turn on • MINUS—turn off • OK/CAL—save current setting and jump to next setting menu
Beeper? :OFF	Set whether to hear the beep sounds while pressing the buttons	• SET/EXIT—exit the setting menu • PLUS—turn on • MINUS—turn off • OK/CAL—save current setting and jump to next setting menu
Irreg mains beep :OFF	Set whether to raise beep alarm in case of voltage malfunction	• SET/EXIT—exit the setting menu • PLUS—turn on • MINUS—turn off • OK/CAL—save current setting and jump to next setting menu
Wait time(sec)? :30	Set the delay time (seconds)	• SET/EXIT—exit the setting menu • PLUS—increase value • MINUS—decrease value • OK/CAL—save current setting and jump to next setting menu
Low cutoff? :180	Set the under-voltage limit (volts)	• SET/EXIT—exit the setting menu • PLUS—increase value • MINUS—decrease value • OK/CAL—save current setting and jump to next setting menu
High cutoff? :180	Set the over-voltage limit (volts)	• SET/EXIT—exit the setting menu • PLUS—increase value • MINUS—decrease value • OK/CAL—exit settings menu

The default delay time for the unit is set to 30 seconds. LCD1 will be continuously updated with the latest mains voltage and room temperature

and the microcontroller will continuously monitor the mains voltage against any over voltage or under voltage. If any excess is detected, the load

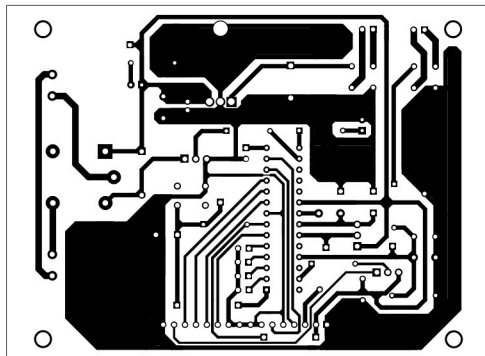


Fig. 5: An actual-size, single-side PCB for the safety timer for home appliances

TABLE II
Test Points

Test point	Details
TP0	0V, GND
TP1	0V-5V depending on AC input (with respect to TP0)
TP2	5V (with respect to TP0)
TP3	10mV/°C
TP4	Low when S5 is pressed

is switched off and the buzzer sounds to indicate irregular voltage. The load will stay in off state for the preset delay time and switch back on once the delay time elapses.

The device settings such as delay time, over-voltage and under-voltage limits can be accessed by pressing the SET/EXIT button. The same button can be used to exit from the settings menu anytime. PLUS (+) and MINUS (-) are used to alter the current setting values. OK/CAL is used to save the settings for the particular setting menu.

Press SET/EXIT to access the device settings. Table I describes the sequential setting menus and function of switches.

Software

The software program is written in BASIC language and compiled with BASCOM-AVR compiler. It is important to note that the code is developed to run on 1MHz frequency and the internal 1MHz RC internal oscillator is used here. The program is burnt in the microcontroller using a suitable programmer with FUSE BYTE settings mentioned below:

Lfuse=0xE1

Hfuse=0xD9

Burn 'digitimer.hex' file in flash memory section and 'digitimer.eep' file in EEPROM section of the microcontroller which can be downloaded from EFY DVD or source.efymag.com.

Construction and testing

An actual-size, single-side PCB recommended for the safety timer for home appliances is shown in Fig. 5 and its component layout in Fig. 6. Assemble the circuit on the PCB to minimise as-